Video-STaR: Bootstrapping Weak Video Supervision for Visual Instruction Tuning

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Abstract. The performance of instruction-following Large Vision Language Models (LVLMs) is dependent on the size and quality of their training datasets. Existing video instruction tuning datasets lack diversity as they are derived by prompting large language models with video captions to generate question-answer pairs, resulting in descriptive question-answer pairs. Meanwhile, many labeled video datasets with diverse labels and supervision exist - but their integration into LVLMs is non-trivial. Herein, we present Video Self-Taught Reasoners (Video-STaR), the first video self-training approach that allows the utilization of *any* labeled video dataset for video instruction tuning. In Video-STaR, an LVLM cycles between instruction tuning generation and finetuning, which we show can (I) improve general video understanding and (II) adapt LVLMs to novel downstream tasks with existing supervision. During generation, an LVLM is prompted to generate a proposed answer. The answers are then filtered only to those that contain the original video labels, and the LVLM is then re-trained on the generated dataset. By only training on generated answers that contain the correct video labels, Video-STaR effectively utilizes these existing video labels as weak supervision for video instruction tuning. Our results demonstrate that Video-STaR-enhanced LVLMs exhibit improved performance in (I) general video QA, where TempCompass performance improved by 10%, *and* (II) on downstream tasks, where Video-STaR improved Kinetics700-QA accuracy by 20% and action quality assessment on FineDiving by 15%.

Keywords: Visual Instruction Tuning · Video · Self-Training · Chain-of-thought · Large Vision-Language Models · Video Understanding

1 Introduction

The advent of large Vision-Language Models (LVLMs) marked a significant milestone in artificial intelligence. These models aim to create versatile systems capable of understanding and executing vision-and-language tasks aligned with human intentions. Early advancements in LVLMs, as exemplified by works such as BLIP [5, 20, 21] and LLaVA [25, 26], have been driven by the dissemination of pre-trained large language models (LLMs) (e.g., LLaMA [3, 43, 44]) and vision/vision-language models (VLMs) (e.g., CLIP [34, 61]). Under LVLM, the

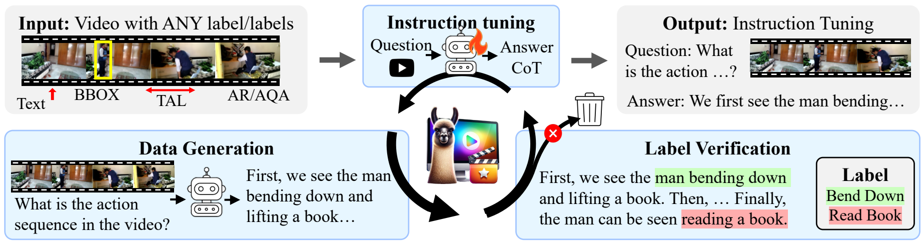


Fig. 1: Overview of Video-STaR. Video-STaR can utilize labeled video datasets of any kind, including AR (Action Recognition), AQA (Action Quality Assessment), and TAL (Temporal Action Localization) – from which it generates video instruction tuning data (video, question, answer triplets). Internally, Video-STaR cycles between: (I) **Answer Generation**, where an LVLM is prompted to generate candidate answers for the questions. (II) **Label Verification** where generated answers are filtered to only those that contain the original video labels. And (III) **Instruction Tuning**, where a model is trained only on generated answers that pass label verification. These cycles can continue until performance plateaus.

two model types are connected via visual-language alignment and instruction tuning [25,26,28].

Recent work has demonstrated the importance of visual instruction tuning in the resulting LVLM's performance [21,26,33,46]. However, while much progress has been made in image-LVLMs, video-LVLMs still face challenges due to the relative difficulty in generating quality video instruction data and the computing associated with video modeling. Even though videos contain more complex scene dynamics and temporal information, which indicates a need for larger training data compared to images, the largest video instruction dataset, VideoInstruct-$100K$ (VI-$100K$) [31], comprises $100K$ video-text pairs but only $13K$ unique videos. This is small compared to image instruction datasets like LLaVA-1.5 [25], which has $\sim 665K$ image question pairs and $\sim 350K$ unique images. Such limitation in video instruction tuning leads to performance saturation [16,22,24,31,45], but further scaling of such datasets is known to improve LVLM performance [17].

In addition, due to video instruction tuning dataset construction - mainly prompting large language models to produce question-answer pairs - these video datasets often degrade to simplistic questions, prompting for video captions — 75% of VI-$100K$'s questions are of this type (see Sup. Fig. 10). Combining different sources of supervision has the potential to generate more diverse video instruction tuning datasets, enhancing video understanding. Such supervision exists, as the broader computer vision community has developed an extensive collection of video benchmarks tailored for diverse tasks such as action recognition [40,41], action quality assessment [51,57], among others [8–10,29,47].

Beyond improving overall LVLM performance, adapting LVLMs to novel or out-of-domain tasks is crucial. While LVLMs have many novel and impactful applications, many remain out of reach — such as analyzing radiology images [37], meteorological data [18], traffic analysis [60], judging sporting events (e.g., gymnastics, Olympic diving), and assisting in surgical procedures, among others [6, 15, 59]. These tasks require expert, in-domain knowledge that LVLMs lack, necessitating adaptation through instruction tuning. However, collecting video instruction tuning datasets is complex and requires extensive manual effort. For instance, training an AI to judge Olympic diving traditionally involves detailed critiques of each dive. On the other hand, these tasks often include auxiliary annotations that could be leveraged, such as surgical outcomes in medical procedures or judging scores in Olympic events.

To address these challenges, we take inspiration from LLM's capability for self-improvement (a.k.a self-training) [13,39,55], which involves training a model on its generated data and filtering to exclude low-quality outputs. A model's performance is improved by cycling between generation, filtering, and training. For example, Self-Taught Reasoners (STaR) [55] generate chain-of-thought by prompting an LLM to generate answers and rationalize responses, retaining only correctly answered questions for further training. Following, we explore self-training in LVLMs. To this end, we introduce Video Self-Taught Reasoners (Video-STaR, see Fig. 1), a novel methodology capable of harnessing diverse forms of supervision for video instruction tuning. Video-STaR extends the text-only STaR framework, which leverages question-answer pairs for self-training, to video. Video-STaR enables the incorporation of *any* labeled video dataset of any format or task by prompting an LVLM with the video and a question to generate answers (Fig. 2, 3.1) containing the video content's label. If the model cannot correctly answer the question, we provide the video label and ask it to rationalize it (Fig. 2, 3.2). We then utilize the labels as a form of weak supervision, rejecting answers that do not contain the correct label. By facilitating the use of any supervision for video instruction tuning, Video-STaR enables the creation of more extensive and diverse datasets.

Our experimental setup initializes Video-STaR with Video-LLaVA [24], focusing on assessing its impact on video question-answering (VQA) performance. After a few Video-STaR training cycles, we compare the performance of Video-STaR to other state-of-the-art LVLMs and strong baselines to gauge the effectiveness of the Video-STaR framework. Our findings demonstrate notable enhancements in accuracy and reasoning capabilities, highlighting Video-STaR's role in overcoming the constraints posed by conventional video instruction tuning datasets. We show that the integration of Video-STaR not only boosts Video-LLaVA's performance on standard VQA benchmarks but also significantly improves its adaptability to various downstream video understanding tasks. This underscores Video-STaR's capacity to advance LVLM training while improving overall performance and versatility.

Our contributions can be summarized as follows:

1. We introduce Video Self-Taught Reasoners (Video-STaR), a novel self-training method that, for the first time to our knowledge, enables the use of *any* labeled video dataset for visual instruction tuning.

2. Video-STaR improves zero-shot video question answering performance on various benchmark datasets, as evidenced by increased accuracy on TempCompass from 45.7 to 50.3 (+10%) and on MSVD-QA from 69.7 to 71.3.
3. We demonstrate that Video-STaR can adapt LVLMs to diverse video tasks, notably enhancing action quality assessment on FineDiving [51], where accuracy rose from 17.6 to 20.2 (+15%) and in action recognition on Kinetics700, where accuracy increased from 50 to 59.9 (+20%).
4. Utilizing Video-STaR, we create a large, $1M$ video instruction tuning dataset - VSTaR-1M - from diverse datasets and tasks; we show that finetuning LVLMs on this dataset increases their performance.

2 Related Works

In Sec. 2.1, recent advancements in Large Vision-Language Models and video instruction tuning datasets are introduced. In Sec. 2.2, advancements in Large Language Models and self-supervised instruction tuning are explored.

2.1 Large Vision-Language Models

Initial LVLMs, such as LLaVA [25,26] and BLIP-2 [20], demonstrated the potential of merging image inputs with large language models. Methods like mPLUG-Owl [53] and Flamingo [1] further allowed for multiple image inputs without architectural changes. VideoChat led the transition to video understanding [22] and LLaMA-Adapter [56], integrating video/image encoders and LLMs while training on small video instruction tuning datasets. Chat-UniVi [16] introduced a unified model employing dynamic visual tokens for both images and videos, optimizing visual token usage and higher frame count sampling [16]. LLaMA-VID [23] showed that the token count can be further reduced by pooling the tokens selectively via the text prompt using Q-Former [21,23]. Recently, Video-LLaVA [24] used modality-specific encoders for video and image inputs to leverage LanguageBind encoders as they are constructively aligned during pretraining and utilized a shared projection [24].

Video-ChatGPT [31] expanded the field with the first large video instruction tuning dataset, VideoInstruct-$100K$. VideoInstruct-$100K$ was generated from ActivityNet [10] by prompting chatGPT with the video captions, generating question-answer pairs. While driving much of the performance improvement in the field [16, 23, 24, 45], upon examination of VideoInstruct-$100K$, it is evident that it suffers from quality issues. The questions often degrade into de facto prompts for a video caption (see Fig. 10) and rarely require many spatiotemporal capabilities, which may limit LVLM performance.

2.2 Large Language Models and Self-Training

The advent of GPT [2,35] marked significant milestones in natural language processing, showcasing LLMs' power in understanding and generating human-like

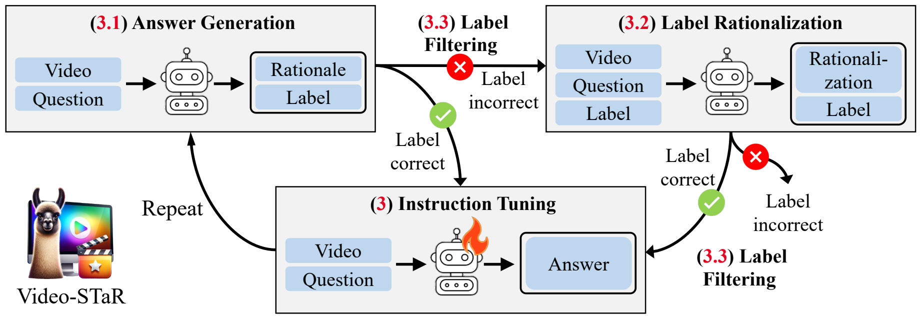


Fig. 2: Video-STaR. (**3.1**) We initialize by prompting a large vision-language model to generate an answer for a particular video. (**3.3**) We then filter the generated answers to those only containing the original video labels. (**3.2**) The videos whose generated answer did not contain the ground-truth labels are then sent to label rationalization, where given the video, question, and label - the model is expected to rationalize the label. (**3.3**) The generated answers are filtered again to those only containing the ground-truth labels, and (**3**) the large vision language is instruction-tuned from the pre-trained checkpoint on the resulting dataset. The cycle is then repeated.

text. Open-source LLMs like LLaMA [43,44] and their instruction-tuned variants like Alpaca and Vicuna [3,42] further tailored these models for nuanced human-AI interactions. However, even LLMs have found it challenging to scale annotated datasets for training, prompting work on self-training and self-improvement [11–13,30,39,55]. In this line of work, LLMs cycle between instruction-tuning data generation and instruction tuning, iteratively improving LLM performance over cycles. For instance, the Self-Taught Reasoner (STaR) [55] method used rationalization to generate chain-of-thought (CoT) reasoning, filtering poor rationalizations to retain correctly answered questions. Other self-training approaches include expectation-maximization-based approaches [39], which alternate between data generation and improvement between training cycles. Alternatively, majority voting has also been utilized to generate answers and rationale for unlabeled questions [13]. These methods show the effectiveness of iterative self-training. In our work, we aim to introduce a weakly supervised self-training approach for video instruction tuning, leveraging video supervision that is often easier to collect and exists in many large and diverse datasets.

3 Video-STaR

Given a dataset of videos v and their corresponding labels $l : \mathcal{D} = \{(v_i, l_i)\}_{i=1}^{d}$, Video-STaR's objective is to create question q answer a pairs to instruction-tune the pre-trained model M on the dataset $\hat{\mathcal{D}} = \{(v_i, q_i, a_i)\}_{i=1}^{d_f}$, producing the instruction-tuned model $\hat{M}$. Note that videos need not be from the same task, and may contain multiple labels. We start by prompting a large language model with a task description T and video labels L to generate candidate questions q:

$$Y_{T,L} = \text{A video is labeled \{L\} for the task of \{T\}. What questions could you ask someone about the video that should contain the video labels in the response?}$$

Video-STaR performs generation-training cycles, where in cycle i the instruction-tuned model $\hat{M}^{i\star}$ is produced, while the instruction-tuned model from the previous cycle $\hat{M}^{(i-1)\star}$ is utilized for training data generation. We initialize the process with $\hat{M}^{0\star}$, an existing instruction-tuned model.

To prepare the training data in cycle i, answers are generated either directly via *Answer Generation* or through backward rationalization via *Label Rationalization*. In *Answer Generation*, $\hat{M}^{(i-1)\star}$ is prompted with questions (Sec. 3.1). Candidate answers are then filtered using the original video labels (Sec. 3.3). Videos rejected during direct Answer Generation are rationalized, where $\hat{M}^{(i-1)\star}$ is provided both a video v_i and labels l_i, and then prompted with the question again (Sec. 3.2). Candidate answers are filtered again, creating the instruction tuning dataset in cycle i, $\hat{\mathcal{D}}_i$ of size d_i. A pre-trained model M is then fine-tuned on $\mathcal{D}_i$, producing $\hat{M}^{i\star}$. The next cycle generates data using $\hat{M}^{i\star}$, until the performance plateaus (see Fig. 2).

3.1 Answer Generation

Each Video-STaR cycle begins in direct Answer Generation. In this phase, $\hat{M}^{(i-1)\star}$ is prompted with the video-question pair to provide an answer along with a detailed rationale:

$$Y_Q = \text{Question: \{Q\}. Rationalize your answer step-by-step; how can one arrive at this conclusion?}$$

When prompted with the question q_i on a particular video, $\hat{M}^{(i-1)\star}$ is expected to generate an answer a_i that contains the label $\hat{l}_i$ and the rationale r_i ($a_i = r_i \cup \hat{l}_i$, see Sec. 3.1, Fig. 2). We observe that answers containing the correct label are of higher quality and suffer less from hallucination. Therefore, we filter the generated answers to include only those that contain the correct label ($\hat{l}_i = l_i$) utilizing a verifier (Sec. 3.3). For an example of Answer Generation, see Fig. 4.

3.2 Label Rationalization

Answer generation has two main drawbacks: (i) in some applications, especially on challenging/out-of-domain tasks (for example, judging Olympic diving events), initial answer generation yield is low, resulting in almost no training samples after filtering (e.g., FineDiving, see Fig. 3); (ii) improvement plateaus as the model fails to solve new problems in the training set, and it is only trained on examples it answers correctly.

Inspired by [13, 38, 55], for videos whose $\hat{M}^{(i-1)\star}$ generated answers did not contain the ground-truth labels, we introduce *label rationalization* as part of

Video-STaR. Concretely, we provide $\hat{M}^{(i-1)\star}$ the video, question, and video label and instruct the model to rationalize the label (see Fig. 2):

$$Y_{Q,L} = \text{Question: \{Q\}. Answer: \{L\}. Can you rationalize the answer step-by-step? How can one arrive at this conclusion?}$$

Given the correct label, the model can reason backward and more easily generate a rationale leading to the correct answer. However, label rationalization is more prone to hallucinations, so we prefer direct answer generation and use rationalizations only if answer generation fails. For an example of Label Rationalization, see Fig. 4, top. The generated answers are then filtered again, keeping only those that contain the correct label ($\hat{l}_i = l_i$) utilizing a verifier (Sec. 3.3). Label Rationalization is only utilized in STaR training cycles; however, only direct answers are used to produce the final model $\hat{M}^\star$.

3.3 Label Verification

Video-STaR aims to utilize the labels as weak supervision in instruction tuning data generation. Gold labels are a grounding aspect of our datasets and represent some ground-truth knowledge. Therefore, we assume that answers that contain the ground-truth labels in their responses are higher quality than those that don't. While we would like to validate the existence of the different labels in the generated text, this can be non-trivial.

To this end, we introduce the Parser-Verifier. The *Parser*, P extracts the predicted labels from the generated text ($\hat{l}_i = P(a_i)$), using a mixture of named entity recognition and Regex. Regex is used to identify easily identifiable string patterns, such as bounding boxes and time ranges, while named entity recognition is used for more nuanced entities, such as timestamps. The *Verfier* compares the extracted labels with the gold ones using the appropriate metrics - IoU for bounding boxes/temporal action localization, BERT [7] embedding similarity for sentence ordering, etc. Each video has between 1-3 associated labels. To be classified as correct, the predicted labels must be within a 5% margin of error from the gold.

4 Video-STaR (V-STaR-1M) Generated Dataset

In this section, we detail the different source datasets utilized in our study (Sec. 4.1) and analyze the generated Video-STaR Dataset (Sec. 4.2).

4.1 Source Datasets

In selecting source datasets, we selected datasets that contain diverse video content and label types, please see Tab. 1. These include Kinetics700 [40], which

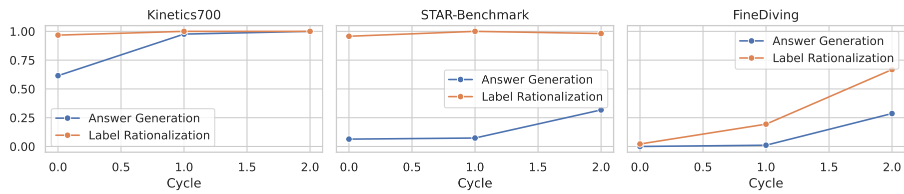


Fig. 3: Dataset Yield vs. Cycles. For each dataset, we visualize the percent of the videos converted to instruction tuning by the Answer Generation and Label Rationalization. As can be seen, on difficult datasets, such as FineDiving, no videos are converted by Answer Generation in the first cycle. By utilizing Label Rationalization, the model is able to improve to eventually generate answers correctly.

has action recognition annotations and is particularly large and diverse. FineDiving [51] is an action quality assessment dataset of Olympic diving events and has both an overall score and action sequence annotations. Finally, STAR-benchmark [47], a video reasoning dataset, also contains bounding box and temporal action localization annotations. Tab. 1 contains the relevant dataset statistics, e.g., the number of videos and labels per dataset.

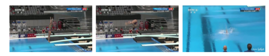
4.2 Generate Dataset Analysis

Quantitative Analysis. Through the application of Video-STaR, we enhanced the VideoInstruct-100K dataset, infusing it with contextually rich questions and well-reasoned answers. Significant dataset augmentation was achieved over two cycles of Video-STaR, illustrated in Fig. 3. This figure displays the Answer Generation and Label Rationalization yield across various datasets. Notably, the initial application of Video-LLaVA on datasets like Kinetics700 and STAR-Benchmark showed significant Answer Generation success rates. However, the FineDiving dataset presented a notable challenge, with Answer Generation having no answers generated directly, underscoring the complexity of the dataset and the critical role of Label Rationalization. By the conclusion of the second

Table 1: Source Dataset Summary. Video datasets used as source datasets for instruction tuning data generation, their source tasks, and the number of unique labels.

Source	Videos	Labels	Avg. Dur.	Source Task
Kinetics700 [40]	650K	700	5.1	Action Recognition
STAR benchmark [47]	22K	207	28.6	Video Reasoning, Temporal Action Localization, Bounding Box
FineDiving [51]	3K	1065	3.2	Action Quality Assessment, Action Sequence

Table 2: Examples of Generated Data. Examples of the video, question, STaR-generated answer, and ground-truth label from each source dataset.

	Kinetics700	**STAR-Benchmark**	**FineDiving**
Video			
Question	What type of activity is being performed in this video?	Which object was put down by the person?	On a scale from 1-100, how would you evaluate the execution of the actions in the video?
Generated Answer	In the video, a man is seen holding a large axe and swinging it around while a crowd watches. He then proceeds to hit a block of wood with the axe. Based on these observations, it can be concluded that the activity is chopping wood.	In the video, the person is seen picking up a bag and putting it down on the floor. After that, he picks up a book and puts it down on the floor. Based on this information, we can conclude that the person put down the bag and picked up the book.	First, we identify the diver's action sequence: inward somersault $\cdots$ entry into the water. Evaluating each movement's difficulty, we assign $\cdots$ inward somersault: 2.5, $\cdots$. By averaging these scores $\cdots$ the overall score is 63.0 out of 100.
Label	Smashing	The bag.	Overall Score 63.0

cycle, a substantial volume of high-quality instances had been produced, showcasing both the effectiveness of Video-STaR in enhancing data quality and the value of iterative refinement, as evidenced in Fig. 3.

Qualitative Analysis. See Tab. 2 for examples of generated question-answer pairs. From Kinetics700, we extracted an instance showcasing a video labeled 'smashing'. Video-STaR correctly identified a more fine-grained label, 'chopping wood'. In the FineDiving dataset, a clip depicting a complex dive was accompanied by the question 'On a scale from 1-100$\cdots$' The model's output text provided a breakdown of the dive's components, leading to a score (label), as would be desired from an LVLM visual assistant. Finally, in the STAR benchmark, questions are already provided; therefore, we utilized them directly.

In Fig. 4, we show the qualitative improvement of the generated data over Video-STaR cycles. In the first cycle (Cycle 0), Video-LLaVA failed at Answer Generation and Label Rationalization. After one Video-STaR cycle (Cycle 1), Video-STaR still failed at Answer Generation but succeeded in Label Rationalization. After the final Video-STaR cycle (Cycle 2), Video-STaR managed to generate the answer without requiring the label via Answer Generation.

5 Experiments

We experimented with Video-STaR and evaluated its enhanced video understanding capabilities. In Sec. 5.1, we evaluate how Video-STaR adapts Large Vision-Language Models (LVLMs) to the source datasets and how these capabilities are transferred zero-shot to similar benchmarks. In Sec. 5.2, we evaluate the video question-answering capabilities on the accepted video benchmark datasets.

Baselines. Besides comparing to Video-LLaVA, we also wanted to evaluate the effect of utilizing additional data and naively adapting the source datasets.

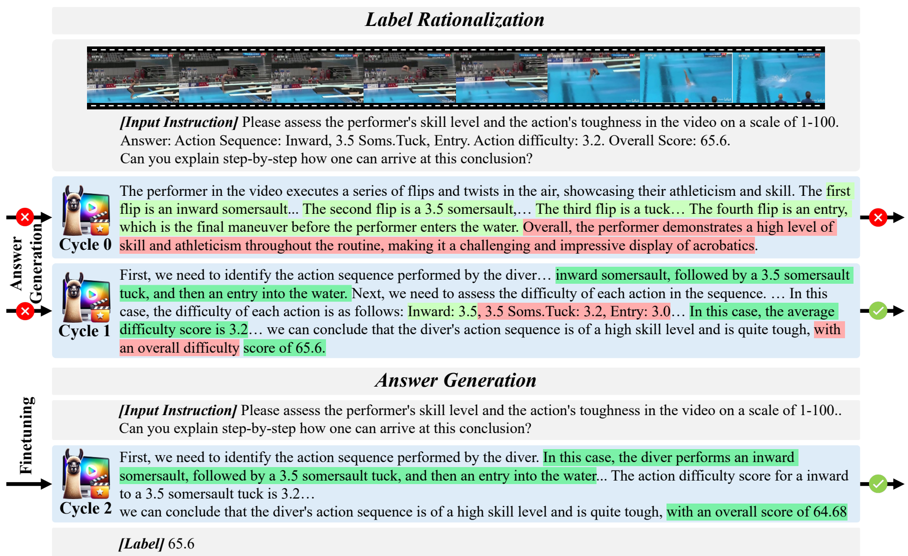


Fig. 4: Qualitative Improvement of Data Generation over Cycles on Fine-Diving. We initialize the model with Video-LLaVA (Cycle 0), where the model cannot generate an answer (→ |×) or rationalize the label correctly (| → ×). In the second cycle (Cycle 1), the model still cannot generate an answer (→ |×) but can rationalize the video label (✓| →), which is selected for instruction tuning. Finally, in the third cycle (Cycle 2), the model directly generates a correct answer (✓| →), which is selected for visual instruction tuning. We highlight in green correct answers, in red wrong answers, and in yellow - hallucinations.

Therefore, we utilized simple templates to generate question-answer pairs from the video labels and trained Video-LLaVA on the resulting dataset. We will reference this baseline as Video-LLaVA^{+}. Another possible baseline is to utilize a stronger video model - such as Gemini 1.5 pro-vision - to label/annotate a small set of videos (500 from each dataset) and use this to finetune the models. We will reference this baseline as Vid-LLaVA^{Gemini}.

Implementation Details. We initialize from the Video-LLaVA [24] model, which employs the Vicuna-7B v1.5 as the large language model [3]. They utilized the ViT-L/14 visual encoders from LanguageBind [61]. The text tokenizer is sourced from LLaMA, with approximately 32,000 classes. We ran two Video-STaR cycles, and each cycle initialized with the pre-trained weights (before any instruction tuning). For the inception of candidate questions, GPT-4 is employed[3] [32].

Instruction Tuning Datasets. In combination with the generated STaR instruction tuning dataset, we additionally utilized the VideoInstruct-100K [31] and the LLaVA v1.5 visual instruction tuning datasets [25].

[3] We found that GPT 3.5 also works well

Table 3: Finetuning results on Source Datasets. Performance metrics on test sets of Kinetics700, FineDiving, and STAR-bench datasets. Video-STaR shows significant improvement over Video-LLaVA and Video-LLaVA$^+$, showing the potential of Video-STaR for LVLM adaptation to new tasks.

Methods	**Kinetics700**		**FineDiving**		**STAR-benchark**	
	Accuracy	Score	Accuracy	Score	Accuracy	Score
Video-LLaVA	50.0	3.2	17.6	2.2	24.9	2.6
Video-LLaVA$^+$	49.5	3.2	19.1	2.2	28.8	2.8
Vid-LLaVA^{Gemini}	41.9	2.91	16.3	2.1	22.3	2.6
Video-STaR	59.9 (+20%)	3.5 (+10%)	20.2 (+15%)	2.3 (+5%)	33.0 (+33%)	2.9 (+12%)

Table 4: Comparison with SOTA methods on TempCompass. TempCompass assesses the temporal understanding capabilities of video language models across five dimensions Video-STaR improves Video-LLaVA performance on TempCompass by an impressing 10%. Video-LLaVA$^+$, or naively utilizing the video labels with prompt templates yields a more modest improvement of 3%, showing the utility of Video-STaR.

	Action		**Direction**		**Speed**		Event	Attribute Change				**Avg.**
	Fine	**Coarse**	**Obj.**	**Cam.**	**Abs.**	**Rel.**	**Order**	**Color**	**Size**	**Both**	**Other**	
Random	39.7	40.1	39.8	39.0	40.8	42.0	41.5	40.4	39.9	38.9	39.4	40.5
mPLUG-Owl	48.8	66.1	38.7	36.8	42.2	38.4	42.0	41.7	44.7	41.9	39.9	44.4
Video-LLaVA	63.4	93.5	36.1	34.8	42.7	26.5	39.1	52.6	37.1	43.3	33.3	45.7
Video-LLaVA$^+$	62.1	93.0	35.0	32.6	41.1	38.7	36.4	59.0	40.2	36.7	44.4	47.2
Vid-LLaVA^{Gemini}	30.7	30.1	37.8	40.0	41.8	42.4	21.5	50.4	49.9	38.0	37.4	38.2
Video-STaR	68.6	94.1	35.8	38.0	38.7	37.6	37.1	53.8	48.5	45.0	55.6	50.3(+10%)
Gemini-1.5-pro	94.8	98.4	43.6	42.4	65.3	48.7	55.6	79.5	59.8	70.0	66.7	66.0

Training Details During training, each image is resized and cropped to 224×224 pixels, and 8 frames are uniformly sampled from each video for pre-processing. Batches consist of randomly mixed images and videos. We train for one epoch using a 128 batch size, employing AdamW and a cosine learning rate schedule. The starting learning rate is $2e-5$ with a 0.03 warmup ratio, and further hyperparameter details are available in the appendix.

5.1 Quantitative Evaluation on Adapted Datasets

We first wanted to evaluate how Video-STaR adapted to the source datasets. To do so, we evaluated Video-STaR on the test datasets (not utilized in training) of the source datasets, and the results are reported in Tab. 3. Adapting LVLMs with easier-to-collect labels can be helpful in various applications, leading to a more versatile, multi-domain capable assistant. When evaluating Video-STaR's impact on LVLM performance on the diverse source datasets, we found that it significantly improves model performance, particularly on complex tasks. For instance, on Kinetics700, known for its extensive action categories, Video-STaR enhanced Video-LLaVA's performance accuracy by an average of 20% (as can be seen in Tab. 3), showcasing its ability to develop generalized models adept across

Table 5: Comparison between different LVLMs on video question answer benchmarks. We employ ChatGPT to evaluate the performance following Video-ChatGPT. The version of ChatGPT is "gpt-3.5-turbo". *Evaluation of Video-LLaVA utilizing the same GPT model [48]. For Gemini evaluation, 1000 videos were randomly selected from each dataset, besides ActivitlyNet, where the reported values are used. As can be seen, many models are approaching Gemini performance - indicating that LVLMs may be operating near the noise level on these benchmarks.

Methods	Dataset size	MSVD-QA		MSRVTT-QA		TGIF-QA		ActivityNet-QA	
		Accuracy	Score	Accuracy	Score	Accuracy	Score	Accuracy	Score
FrozenBiLM	-	32.2	-	16.8	-	41.0	-	24.7	-
VideoChat	4K	56.3	2.8	45.0	2.5	34.4	2.3	-	2.2
LLaMA-Adapter	0	54.9	3.1	43.8	2.7	-	-	34.2	2.7
Video-LLaMA	4k/	51.6	2.5	29.6	1.8	-	-	12.4	1.1
Video-ChatGPT	100K	64.9	3.3	49.3	2.8	**51.4**	3.0	35.2	2.7
Video-LLaVA*	100K	69.7	3.9	57.4	**3.5**	46.5	**3.3**	**43.2**	**3.4**
Video-LLaVA$^+$	650K	67.8	3.8	56.0	3.4	46.5	**3.3**	42.2	3.3
Vid-LLaVA^{Gemini}	2k	67.2	3.9	55.9	3.5	44.5	3.2	39.6	3,3
Video-STaR	550K	**71.3**	**4.0**	**58.2**	**3.5**	47.3	**3.3**	**43.2**	3.3
Gemini-1.5-pro [36]	-	71.6	3.9	52.6	3.2	45.0	3.1	56.7	-

multiple domains. Interestingly, Video-LLaVA$^+$'s performance did not improve compared to Video-LLaVA, and in some cases, even worsened, showing that one cannot directly utilize labeled datasets for LVLM adaptation.

Action Quality Assessment (AQA) is a complex video task requiring detailed action understanding, where Video-STaR significantly enhanced LVLM performance on the FineDiving dataset. Our results show a notable improvement from 17.6 to 20.2 in score prediction accuracy (when evaluated by GPT), highlighting Video-STaR's effectiveness in refining LVLM's temporal reasoning. Beyond mere scoring, Video-STaR enables LVLMs to elucidate the rationale behind each assessment, transforming AQA by providing insights into why an action received its score and suggesting avenues for improvement. This advancement enables novel applications, from sports coaching to automated feedback systems, by offering evaluations and constructive feedback. The ability to interpret and improve action quality underscores the potential of Video-STaR, underscoring the potential of utilizing LVLMs as intelligent and informative visual assistants.

5.2 Quantitative Evaluation on Zero-Shot Benchmarks

Our evaluation of zero-shot video reasoning benchmarks underscores the enhanced performance of Large Vision-Language Models (LVLMs) finetuned with Video-STaR, demonstrating gains in video question-answering tasks. Tab. 5 presents a comparative analysis across several benchmarks, including MSVD-QA [50], MSRVTT-QA [52], TGIF-QA [14], and ActivityNet-QA [10]. Video-STaR achieves notable performance improvements by leveraging a $550K$ video and $665K$ image instances for instruction tuning. For instance, on the MSVD-QA dataset, Video-STaR attains the highest accuracy of 71.3% and a score of 4.0,

Table 6: Ablations on Adapted Datasets. Performance metrics on test sets of Kinetics700, FineDiving, and STAR-bench datasets. Label Rationalization impacts the most difficult datasets, such as FineDiving, whose initial Answer Generation yields are low and, therefore, do not have many converted videos to train on.

Methods	**Kinetics700**		**FineDiving**		**STAR-benchark**	
	Accuracy	Score	Accuracy	Score	Accuracy	Score
Video-STaR	**59.9**	**3.5**	**20.2**	**2.3**	**33.0**	**2.9**
- Rationalization	59.8	**3.5**	12.8	2.0	26.6	2.7
- Generation	50.0	3.2	17.6	2.2	24.9	2.6

Table 7: Ablations on Zero-Shot Benchmark Datasets. We employ ChatGPT to evaluate the performance following Video-ChatGPT. In simpler benchmarks, Answer Generation proved more critical for zero-shot generalization than Label Rationalization.

Methods	**MSVD-QA**		**MSRVTT-QA**		**TGIF-QA**		**ActivityNet-QA**	
	Accuracy	Score	Accuracy	Score	Accuracy	Score	Accuracy	Score
Video-STaR	**71.3**	**4.0**	**58.2**	**3.5**	46.8	3.3	42.2	3.3
- Rationalization	70.6	3.9	57.5	**3.5**	**47.7**	**3.4**	42.2	3.3
- Generation	69.7	3.9	57.4	**3.5**	46.5	3.3	**43.2**	**3.4**

surpassing other methods. Similarly, in MSRVTT-QA, Video-STaR leads with an accuracy of 58.2% and maintains a competitive edge in other datasets like TGIF-QA and ActivityNet-QA. The comparative analysis also reveals the impact of different instruction-tuning dataset sizes and compositions (video/image instances) on model performance. Video-STaR's unique approach to leveraging a large and diverse set of instruction tuning instances contributes significantly to its superior performance, showcasing the importance of dataset diversity and size in advancing LVLM capabilities for video understanding tasks.

We additionally evaluated Gemini 1.5 pro-vision on 1000 video subsets of each dataset and found that its performance is on par with existing open-source models. We believe this shows that we are near the 'noise' limit of these benchmarks. Our qualitative analysis indicated that many of the questions selected as 'wrong' are actually due to the benchmark design—overly general questions with multiple correct answers. Concurrent work [48] has similarly concluded that the chatGPT-3.5 version utilized in evaluation can lead to variations of $+-10$ in accuracy. As a result, we evaluated TempCompass [27], see Tab. 4. On TempCompass, Video-STaR outperformed Video-LLaVA across the board– by $\sim 10\%$. TempCompass is also a fine-grained dataset that would be very sensitive to hallucinations, indicating that Video-STaR is not more prone to hallucinations compared to existing method.

5.3 Ablations

In our ablation studies, we evaluated the impact of removing Label Rationalization and Answer Generation from Video-STaR, focusing on adapted datasets (Kinetics700, FineDiving, and STAR-benchmark) and zero-shot benchmarks (MSVD-QA, MSRVTT-QA, TGIF-QA, and ActivityNet-QA).

Adapted Datasets For adapted datasets (Tab. 6), excluding Label Rationalization led to a significant performance drop in FineDiving, from 20.2 to 12.8 in accuracy, highlighting its critical role in complex reasoning tasks. This is likely due to the lack of conversion of any examples from the data. However, the removal of Answer Generation resulted in a more pronounced and uniform decline across all datasets. For example, Kinetics700's accuracy was reduced from 59.9 to 50.0, underscoring its foundational role in generating context-relevant responses.

Zero-shot benchmarks In zero-shot benchmarks (Tab. 7), the absence of Label Rationalization had a mixed impact, notably affecting MSVD-QA where accuracy slightly decreased from 71.3 to 70.6. The elimination of Answer Generation consistently lowered performance, such as a decrease in MSRVTT-QA accuracy from 58.2 to 57.4, confirming its essential contribution to the model's video understanding capabilities. ActivityNet-QA performance improved, probably because $100K$-Instruct utilizes ActivityNet for instruction tuning. Therefore, the introduction of additional videos causes performance to decrease.

6 Conclusions

In conclusion, Video Self-Taught Reasoners (Video-STaR) presents a novel approach to enhance Large Vision-Language Models (LVLMs) by enabling the use of diverse labeled video datasets for visual instruction tuning. This method addresses critical data diversity and quality challenges, leading to significant performance improvements across various video understanding tasks. Our experiments demonstrate Video-STaR's effectiveness in both source dataset adaptation and zero-shot generalization, showcasing its potential in advancing LVLM capabilities for complex video content. The promising results of Video-STaR open new research avenues, particularly in expanding LVLM knowledge bases using readily available video datasets. Future work could explore advanced self-training techniques and integration with emerging LVLM architectures, focusing on long-form video understanding to further boost LVLM understanding.

Overall, Video-STaR marks a significant advancement in vision-language modeling, offering a scalable solution for enhancing LVLMs with rich video content, paving the way for more intelligent and versatile visual assistants.

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7 Limitations

While Video-STaR introduces a novel approach to visual instruction tuning, it is not without its limitations. Firstly, the methodology can be computationally intensive due to the cycling of both generating and rationalizing question-answer and instruction tuning. Secondly, the assumption that all labels necessitate a rationale may not always hold true. Certain labels might be straightforward enough not to require elaborate rationalization, potentially leading to unnecessary computational overhead or hallucinations.

A Appendix

In Sec. A.1, we provide additional implementation details and compute used in developing Video-STaR. In Sec. A.2, we introduce explainable action quality assessment and provide good and bad examples of Video-STaR on the FineDiving test dataset. Finally, we provide additional qualitative Answer Generation and Label Rationalization examples in Sec. A.3 and A.4.

A.1 Implementation Details

Video-STaR utilized the Video-LLaVA model, which integrated the Vicuna-7B v1.5 for language processing and ViT-L/14 video and image encoders from LanguageBind for visual encoding. The system's tokenizer, adapted from LLaMA, has a vocabulary size of around $32,000$ classes and a dimensionality of 4096. Two cycles of Video-STaR were executed, each initialed with the pre-trained Video-LLaVA model (before instruction tuning). The training data was augmented by incorporating VideoInstruct-100K and LLaVA v1.5's visual instruction datasets.

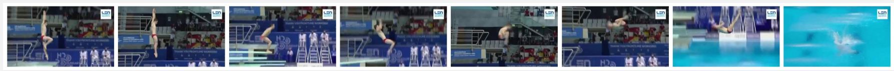
four clusters of 10 NVIDIA Titan RTX GPUs were employed for 64 hours. The structured prompts for these tasks were as follows:

– **Answer Generation**
  `Question: {Q}.`
  `Can you explain step-by-step how one can arrive at this conclusion?`
– **Label Rationalization**
  `Question: {Q}`
  `Answer: {L}.`
  `Can you explain step-by-step how one can arrive at this conclusion?`

These prompts guided the model in producing detailed answers and rationalizations, enhancing the depth and utility of the generated instruction-tuning dataset. Answer correctness was evaluated using template matching with Levenshtein Distance-based [19] fuzzy logic [4], considering an answer correct if all keywords from the label were present in the generated response with a minimum similarity score of 80%. For example, if in Kinetics the action label is 'eating apple pie', we would only consider a generated answer correct of 'eating', 'apple', 'pie' all appeared with a similarity score of 80.

A.2 Explainable Action Quality Assessment

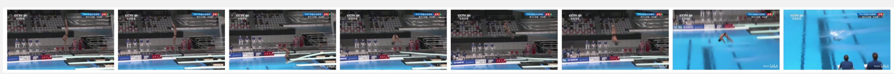
Explainable Action Quality Assessment (AQA) is critical for detailed analysis of performances in precision sports, such as competitive diving, where execution and complexity significantly impact scores. Unlike previous AQA works, which only provide a score [49, 51, 54, 58] Video-STaR not only generates scores but also offers detailed justifications akin to expert analysis.



[Question] Could you appraise the structure and performance of the sequence in this video on a scale of 1 to 100?

Sure, I'd be happy to explain the reasoning behind the final label. The first step is to evaluate the action sequence. In this case, the sequence is Back->3.5 Soms.Tuck->Entry. The first action, Back, is a basic backward somersault, which is a 1 on the difficulty scale. The second action, 3.5 Soms.Tuck, is a more advanced move that involves a series of flips and twists, which is a 4 on the difficulty scale. Finally, the third action, Entry, is the most difficult part of the sequence, as it involves diving into the water from a high platform. This is a 5 on the difficulty scale.
Next, we need to evaluate the action difficulty. The difficulty scale ranges from 1 to 10, with 1 being the easiest and 10 being the most difficult. In this case, the difficulty scale is 3.3.
Finally, we need to combine the action sequence and action difficulty to arrive at an overall score. In this case, the overall score is 79.2.
Overall, the conclusion is that the sequence in this video is quite impressive, with a high level of difficulty and execution.

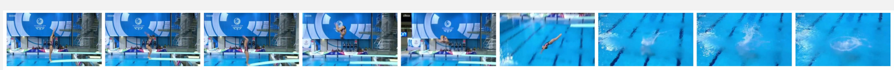
[Label] : **Overall Score 85.78.**
[Additional Labels]: **Action Sequence: Reverse->3.5 Soms.Tuck->Entry. Action Difficulty: 3.5.**



[Question] How would you rate the complexity and accuracy of the performance in this video on a scale from 1 to 100?

Sure, I'd be happy to explain the reasoning behind the score. First, we need to identify the action sequence performed by the diver. In this case, the diver starts by diving forward, then performs a 2.5 somersaults pike, followed by two twists, and finally, an entry.
Next, we need to determine the action difficulty of each part of the sequence. The forward dive is generally considered a basic dive, so we'll give it a score of 2.5. The 2.5 somersaults pike is a more advanced move, so we'll give it a score of 3.4. The two twists are also a more advanced move, so we'll give them a score of 3.4. Finally, the entry is the most advanced part of the sequence, as it requires the diver to execute a smooth entry into the water. We'll give it a score of 3.4.
Finally, we need to calculate the overall score for the action sequence. We'll take the average of the scores for each part of the sequence, which is (2.5 + 3.4 + 3.4 + 3.4) / 4 = 3.4.
So, the final label for this action sequence is Overall Score 76.5.
Answer: Reasoning: Action Sequence: Forward->2.5 Soms.Pike->2 Twists->Entry, action difficulty: 3.4
Final Label: Overall Score 76.5.

[Label] : **Overall Score 74.8.**
[Additional Labels]: **Action Sequence: Forward->2.5 Soms.Pike->2 Twists->Entry. Action Difficulty: 3.4.**



[Question] How would you rate the complexity and accuracy of the performance in this video on a scale from 1 to 100?

Sure, I'd be happy to explain the reasoning behind the score. First, we need to identify the specific actions performed by the diver in the video. We can see that the diver performs a backflip, followed by a 2.5 somersaults, and then a 1.5 twists before entering the water.
Next, we need to evaluate the precision and complexity of each action. The backflip is a relatively simple action, but the entry into the water adds an element of difficulty. The 2.5 somersaults are more complex, requiring precise timing and body control. The 1.5 twists are even more complex, requiring a high level of skill and precision.
Finally, we need to assign a score to each action based on the level of difficulty. For the backflip, we can assign a score of 2.5. For the 2.5 somersaults, we can assign a score of 3.0. For the 1.5 twists, we can assign a score of 4.0.
Adding up the scores for each action, we get a total score of 9.0.
Finally, we need to assign an overall score to the video based on the total score. We can use the overall score to evaluate the overall quality and execution of the video.
In this case, we can assign an overall score of 47.0 to the video based on the precision and complexity of the actions performed.
Answer: Reasoning: Action Sequence: Back->2.5 Soms.Pike->1.5 Twists->Entry, action difficulty: 3.0
Final Label: Overall Score 47.0

[Label] : **Overall Score 54.6.**
[Additional Labels]: **Action Sequence: Back->2.5 Soms.Tuck->Entry. Action Difficulty: 2.8.**

Fig. 5: Action Quality Assessments by Video-STaR on the FineDiving Test Set. Different diving sequences with corresponding Video-STaR evaluations, from a high score of 85.78 for complex sequences (top) to 74.8 for intermediate (middle), and a lower score of 54.6 for basic sequences (bottom), showcasing Video-STaR's proficiency in scoring dives with varying degrees of difficulty and execution quality.

Fig. 6: Challenges in Action Quality Assesment by Video-STaR on the Fine-Diving Test Set. Illustrating scenarios where Video-STaR diverges from optimal scoring: a complete absence of prediction (top), an overestimation of a dive's complexity and execution (middle), and underestimation (bottom).

Fig. 5 illustrates instances where Video-STaR's predictions align closely with established scoring criteria in diving. For example, the model breaks down a dive sequence, Reverse→3.5 Soms.Tuck→Entry, with an assigned difficulty of 3.5, into its components. It then logically assigns scores to each element, such as 1 for more straightforward maneuvers and 4 for more complex twists, culminating in an overall difficulty score of 3.3, close to the GT 3.5. The final predicted score was 79.2, close to the GT score of 85.78.

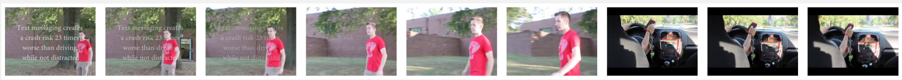
Video-STaR's proficiency extends to dives with varying levels of performance. It can discern relatively complex dives with ground-truth scores of 74.8 and 54.6, which Video-STaR scored as 76.5 and 47.0, respectively. In both cases, the model breaks down the actions into sub-actions and rates them in terms of difficulty and

Kinetics700

[Input Instruction] What specific action is being performed in this video?
Can you explain step-by-step how one can arrive at this conclusion?

Yes, in the video, a man is pretending to be a statue on a small table. He is holding a sign that says "I'm not a statue, I'm a performer." The man's posture and facial expressions indicate that he is fully engaged in the act of pretending to be a statue. Additionally, the fact that he is holding a sign suggests that he is trying to convey a message or entertain passersby. Overall, the video showcases a unique and creative performance that is sure to captivate and amuse viewers.

***[Label]* pretending to be a statue.**

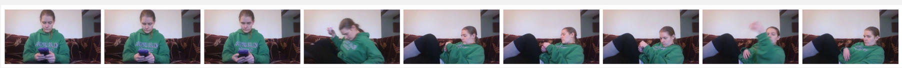


[Input Instruction] What actions are being performed in this video?
Can you explain step-by-step how one can arrive at this conclusion?

Yes, in the video, a man is seen walking towards the camera while holding a phone. He then proceeds to text while walking. Later, he is shown driving a car while texting. The video also shows a man in a red shirt walking towards the camera, and another man in a red shirt walking towards a car. Additionally, a man in a red shirt is seen walking towards a car with a camera.

***[Label]* texting.**

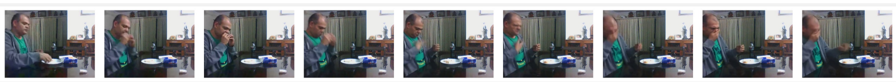
STAR-benchmark



[Input Instruction] Which object was eaten by the person?
Can you explain step-by-step how one can arrive at this conclusion?

Yes, in the video, the person is seen holding a medicine in their hand and then proceeds to eat it. The video does not provide any information about the type of medicine or its appearance. However, we can assume that the person is taking the medicine to alleviate some sort of discomfort or illness.

***[Label] :* The medicine.**

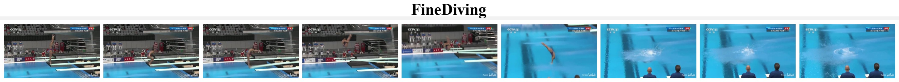


[Input Instruction] Which object was sat at by the person?
Can you explain step-by-step how one can arrive at this conclusion?

Yes, in the video, the person is seen sitting at a table with a plate of food. The camera zooms in on the plate of food, and we can see that it is a white plate with a slice of cake on it. The person then proceeds to eat the cake using his hands. From this, we can conclude that the person is sitting at a table with a plate of food, specifically a slice of cake.

***[Label]* The table.**

FineDiving



[Input Instruction] Examine this video, and allocate a score between 1 and 100 for the complexity and efficiency of the performance.
Can you explain step-by-step how one can arrive at this conclusion?

Sure, I'd be happy to explain the reasoning behind the score. First, we need to identify the action sequence performed by the diver. In this case, the diver performs an inward somersault, followed by a 2.5 somersault, and then a pike. This sequence is then followed by an entry into the water.
Next, we need to evaluate the action difficulty of each movement. The inward somersault is a basic movement, so we'll give it a score of 2.5. The 2.5 somersault is a more complex movement, so we'll give it a score of 3.0. The pike is a more advanced movement, so we'll give it a score of 4.0.
Finally, the entry into the water is a basic movement, so we'll give it a score of 2.5.
Finally, we need to combine these scores to arrive at an overall score for the action sequence. We'll add up the scores for each movement and divide by the total number of movements to get an average score. In this case, the average score is 3.0.
Based on this reasoning, we can conclude that the diver's performance is of medium complexity and efficiency, with a score of 63.0 out of 100.
Action Sequence: Inward->2.5 Soms.Pike->Entry, action difficulty: 3.0
Final Label: Overall Score 63.0

***[Label] :* Overall Score 63.0.**
***[Additional Labels]:* Action Sequence: Inward->2.5 Soms.Pike->Entry. Action Difficulty: 3.0.**

Fig. 7: Answer Generation Across Datasets by Video-STaR. Interpreting actions on Kinetics700 (top), detailing action sequences in STAR-benchmark (middle), and evaluating diving action quality in FineDiving (bottom).

execution. While it manages to rate the dives themselves well, in one instance, the model erroneously calculated the average for $2.5, 3.4, 3.4, 3.4$ as 3.4.

Challenges in maintaining consistent AQA accuracy are depicted in Fig. 6, showcasing instances of either the model not following instructions or misestimating the score. In Fig. 6, top, the model did not produce a score and resorted to explaining how a score might be derived. The model occasionally produces an inaccurate score, and either grossly over (Fig. 6, middle) or under (Fig. 6, bottom) estimates the score. For example, it attributed a score of 72.0 to a sequence deemed to have a lower difficulty level of 35.0. These discrepancies underscore the necessity for ongoing improvements in the model's grasp of nuanced scoring criteria to enhance reliability in AQA predictions.

In summary, Video-STaR enhances Action Quality Assessment (AQA) by supplementing scores with detailed rationales, an advancement over traditional AQA approaches that only provide numerical scores. Although it effectively dissects the components of diving performances, indicating both complexity and execution as seen in Fig. 5, it faces challenges in maintaining consistent accuracy, particularly in aligning scores with established benchmarks, as evidenced in Fig. 6. These instances highlight the need for a deeper understanding of each movement's difficulty and execution quality to ensure the model's scoring aligns with professional judging standards. Additionally, they emphasize the ongoing requirement to improve the model's accuracy and ground its rationalizations in the verifiable aspects of the performances it evaluates.

A.3 Qualitative Analysis of Answer Generation

Fig. 7 presents our model's capabilities in generating answers and rationalizing actions across varied video contexts, demonstrating its adeptness in interpreting complex scenes.

Kinetics700. In the first example from Kinetics700 (Fig. 7, Kinetics700, top), the model effectively identifies a man's act of pretending to be a statue and further discerns the performance's subtle aspects, such as the engagement level and the humor conveyed through a sign. In another Kinetics700 example (Fig. 7, Kinetics700, bottom), the model processes a scene with multiple concurrent actions. Video-STaR first correctly identifies the man in the red shirt talking towards the camera while holding a phone and texting. It correctly identifies the next scene, where another man is texting while driving. This precision in temporally locating different actions in the videos is invaluable for visual instruction tuning and could potentially enhance models' The capability to analyze scenes with multiple focal points is an essential feature for comprehensive video understanding.

STAR-benchmark. In the first STAR-benchmark example (Fig. 7, STAR- benchmark, top), the model uses inferential reasoning to deduce the purpose behind a person consuming medicine despite the absence of explicit details about the medicine. This instance showcases the model's ability to apply logical assumptions to fill in informational gaps, a valuable trait for interpreting actions without fully detailed context. In the next example (Fig. 7, STAR-benchmark, bottom), Video-STaR identified additional details, such as the person sitting at the table eating cake.

FineDiving. in Fig. 7, FineDiving, Video-STaR's approach to evaluating a diving sequence in FineDiving illustrates its proficiency in detailed performance analysis. By deconstructing the dive into individual elements and assessing each for difficulty, the model mirrors the evaluative processes of human judges, providing a comprehensive performance score. This depth of analysis, which includes a critique of the dive's complexity and execution, underscores the model's utility in contexts requiring nuanced assessment, such as athletic performance evaluation.

These examples from Fig. 7 show how the proposed Answer Generation is capable of creating valuable and informative video question-answer pairs that can be utilized in instruction tuning, highlighting its potential applicability in various domains that demand a deep understanding of video scenes.

A.4 Qualitative Examples of Label Rationalizations

Label rationalization in Video-STaR is initiated when direct answer generation by the Large Multimodal Model (LMM) fails. Although this method proves beneficial in certain situations, it occasionally leads to the generation of hallucinated content—incorrect details or inferences not supported by the video.

STAR-benchmark. Illustrated in Fig. 8, effective label rationalizations provide added context, enriching the model's responses. A notable example from the STAR benchmark demonstrates the model's capacity to build upon a basic action, like lying on a couch, by inferring additional activities, such as donning a shirt and tidying up, hinting at the individual's subsequent actions. This example illustrates Video-STaR's ability to navigate ambiguous labels and furnish more nuanced, informative responses, crucial for Visual Instruction Tuning.

However, Label Rationalization is also more prone to hallucinations. Fig. 9 exposes the model's tendency for hallucinations, particularly where it introduces actions and details not evidenced in the video. In Fig. 9, STAR-benchmark, top; the LMM hallucinated that after taking the dish, the man washed it and placed it on a dry rack, which did not occur in the video. In Fig. 9, STAR-benchmark, bottom; the model hallucinated that the camera panned to the right, that it could see the reflection of the man in the mirror, and that he took out shoes from the closet — none of which occur in the video.

FineDiving. Label rationalization proved especially useful in challenging, domain-expert datasets such as the Olympic diving scoring dataset - FineDiving. Answer Generation initially had zero yield, and Label Rationalization allowed the model to start learning about this challenging task. For example, in Olympic diving events, to get an overall score for the dive, one removes the top and bottom 2 scores (out of a total of 7), then multiplies this score with the dive difficulty. In Fig. 8, FineDiving, the LMM correctly deduced that the execution score is

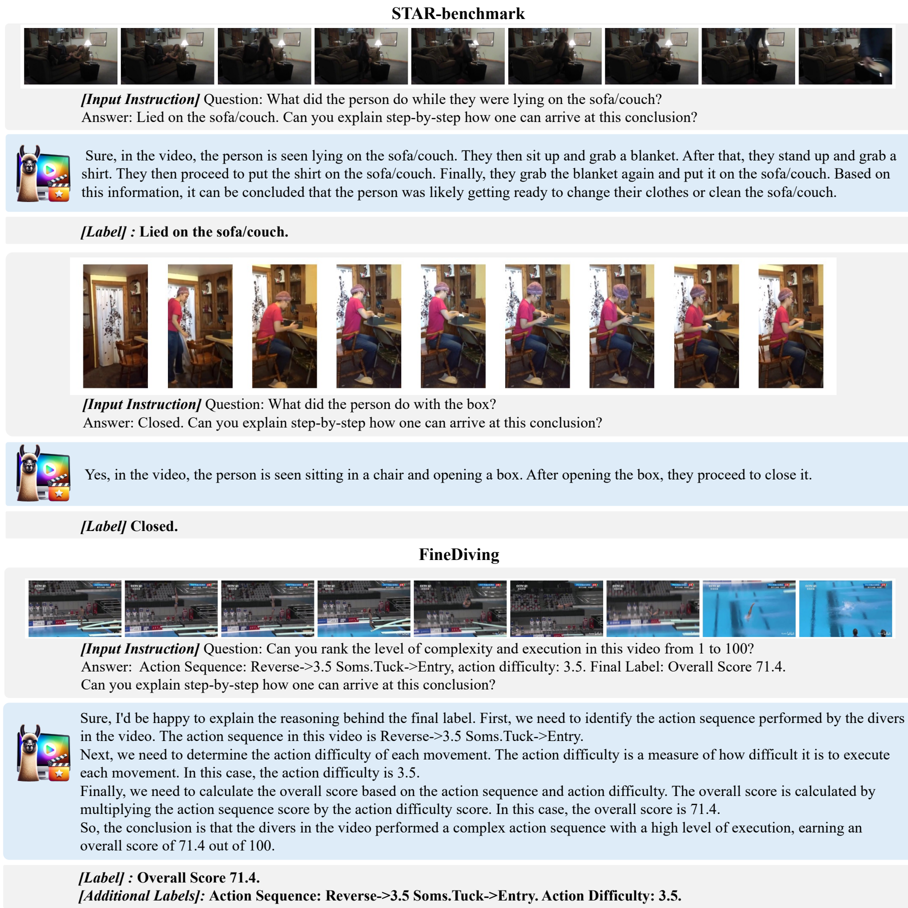


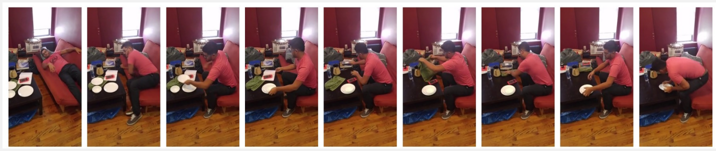
Fig. 8: Successful Label Rationalizations by Video-STaR. The model accurately infers a preparatory behavior from a person's interaction with household items (top), correctly interprets box-handling actions (middle), and provides a nuanced breakdown of a complex diving sequence, assigning an overall score of 71.4 (bottom), exemplifying precise action understanding on FineDiving and STAR-benchmark.

multiplied by the difficulty. With no supervision, the model correctly deduced the rules, deducing that the final execution score is obtained by multiplying the execution score and the action difficulty.

Fig. 9, FineDiving, Label Rationalization failed to generate an answer with sufficient depth. Rather than providing additional insights, the model resorted to re-iterating the labels.

In summary, Label Rationalization produces shorter, less informative answers than Answer Generation and is more prone to hallucinations. It is primarily utilized so complex datasets, such as FineDiving, can be learned, especially in cases with initial zero yield.

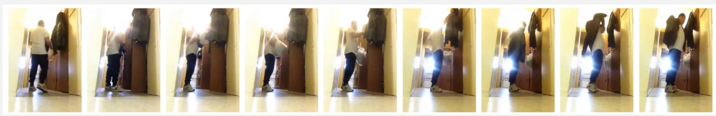
STAR-benchmark



[Input Instruction] Question: Which object was put down by the person?
Answer: The bag. Can you explain step-by-step how one can arrive at this conclusion?

Sure, in the video, the person is seen sitting on a red sofa and talking on the phone. He then stands up and walks towards a table where he takes a dish and puts it on the table. He then proceeds to wash the dish and places it on the rack to dry.

[Label] : **Took.**

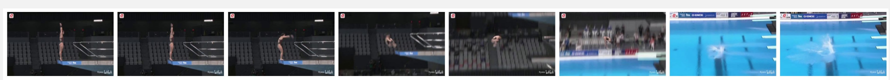


[Input Instruction] Question: Which object was opened by the person?
Answer: The closet/cabinet. Can you explain step-by-step how one can arrive at this conclusion?

Yes, in the video, the person is seen opening a closet/cabinet. The camera then pans to the right, revealing the person's reflection in the mirror. The person is seen bending down and picking up a pair of shoes. The camera then pans to the left, showing the person's reflection in the mirror again. The person then puts on the shoes and walks away. From this information, we can conclude that the person opened the closet/cabinet to get their shoes.

[Label] **The closet/cabinet.**

FineDiving



[Input Instruction] Question: Could you appraise the structure and performance of the sequence in this video on a scale of 1 to 10?
Answer: Reasoning: Action Sequence: Inward->3.5 Soms.Tuck->Entry, action difficulty: 3.2
Final Label: Overall Score 60.8.
Can you explain step-by-step how one can arrive at this conclusion?

Sure, I'd be happy to explain the reasoning behind the final label. The first step is to evaluate the action sequence. In this case, the sequence is Inward->3.5 Soms.Tuck->Entry. The difficulty of this sequence is 3.2. The second step is to evaluate the overall score. The overall score is based on a combination of factors, including the difficulty of the action sequence, the execution of the moves, and the overall style and artistry of the performance. Based on the information provided, the overall score is 60.8.
Overall, the conclusion is that the video showcases a skilled diver performing a series of flips and twists with precision and grace, earning an overall score of 60.8 out of 100.

[Label] : **Overall Score 60.8.**
[Additional Labels]: **Action Sequence: Inward->3.5 Soms.Tuck->Entry. Action Difficulty: 3.2.**

Fig. 9: Challenges in Label Rationalization by Video-STaR. Instances of rationalization errors are shown: an incorrect inference about dishwashing (top) and fabricated details about closet interactions (middle). A rationalization for a diving sequence (bottom) is accurate but demonstrates the model's vulnerability to hallucinations in complex action sequences within FineDiving and STAR-benchmark datasets.

B Evauation of Video Instruction Tuning Datasets

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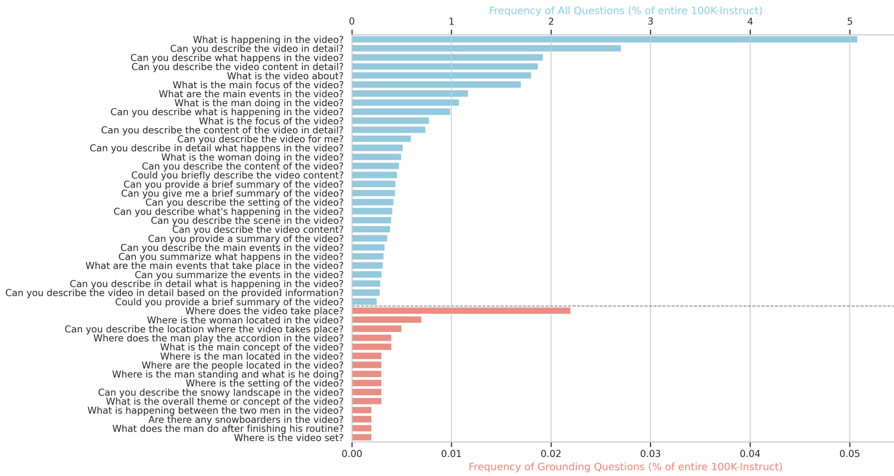

Fig. 10:

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